

The matrix $\begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & 0 \end{bmatrix}$ has no ~~zero~~ leading entry in Row 3.

$$\begin{array}{c} \boxed{\text{Row 2} = \text{Row 2}} \\ \times 1/3 \end{array} \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 5/3 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{c} \boxed{\text{Row 1} = \text{Row 1}} \\ + 2 \times \text{Row 2} \end{array} \rightarrow \begin{bmatrix} 1 & 0 & 7/3 \\ 0 & 1 & 5/3 \\ 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1 and Row 2 are non-zero rows. Row 3 is a zero row.

$\therefore$ The condition that all the non-zero rows must be above the zero rows is True.

2. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions below it must contain 0 is True.

2. Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2

$\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

$\therefore$ The final matrix is in rref. A cannot be reduced to I in rref. By Theorem 7, A has no inverse.

3) If A is an invertible matrix, prove that $5A$ is an invertible matrix.

Solution: By defn, there exists a unique matrix, A^{-1} s.t. $AA^{-1} = A^{-1}A = I$.

$$\begin{aligned} (5A) \left(\frac{1}{5} \cdot A^{-1} \right) &= \frac{1}{5} \cdot (5A \cdot A^{-1}) \quad (\text{Theorem 2(d)}) \\ &= \frac{1}{5} \cdot 5(AA^{-1}) \quad (\text{Theorem 2(d)}) \\ &= I \quad (\text{Given}) \end{aligned}$$